Generic Sequential Convergence of Lloyd Algorithms Without Definability Assumptions: A Partial Resolution

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Abstract

Portales, Cazelles, and Pauwels proved sequential convergence of optimal- and uniform-quantization Lloyd iterations under a global subanalyticity assumption on the target density and asked for convergence mechanisms beyond definability. This note gives a substantial partial resolution, not a complete solution of the unrestricted problem.

First, a deterministic argument shows that a precompact Lloyd orbit converges whenever the critical set on its limiting energy level is totally disconnected. A local Morse–Bott condition also implies convergence, finite length, and an eventual geometric rate. Second, let $\Omega \subset \mathbb{R}^d$ be bounded, open, convex, and Lipschitz, and let

$$\mathcal{P}^1_+(\overline{\Omega}) = \left\{ f \in C^1(\overline{\Omega}) : \min_{\overline{\Omega}} f > 0,\ \int_\Omega f = 1 \right\}.$$

There is a residual set $\mathcal{R} \subset \mathcal{P}^1_+(\overline{\Omega})$ such that, for every $f \in \mathcal{R}$ and every number N of sites, both the optimal and uniform quadratic quantization objectives are Morse on the manifold of distinct sites in Ω. Consequently, every corresponding Lloyd sequence with distinct initialization converges to a nondegenerate critical point, has finite length, and is eventually geometrically convergent.

The genericity proof uses a cellwise zero-mass perturbation of the density. A bump supported strictly inside each Voronoi or Laguerre cell independently prescribes the associated component of the gradient variation. Thus the universal gradient map is a submersion, and finite-dimensional parametric transversality yields the Morse property. The result applies to a residual set of densities that are not globally subanalytic, but it does not cover all absolutely continuous targets or even every positive C^1 density.

1 Introduction and status of the problem

Let $\mu = f(x)\,\mathrm{d}x$ be an absolutely continuous probability measure with compact support. Lloyd's method is a fixed-point algorithm for two related semi-discrete quantization problems: optimal quantization, in which the masses of the output atoms are free, and fixed-weight quantization, in which the masses are prescribed. Portales, Cazelles, and Pauwels proved sequential convergence for the optimal and uniform variants when the density is globally subanalytic; their proof obtains a Kurdyka–Łojasiewicz inequality from definability [5]. They also emphasize that, beyond such rigidity assumptions, a general sequential-convergence guarantee remains open, and they identify second-order analysis of critical points as a natural next step.

The purpose of this note is to develop that second-order route. The main conclusions are:

(i) a purely topological convergence criterion: the critical set on the limiting energy level need only be totally disconnected;

(ii) a local Morse–Bott criterion giving finite length and geometric convergence;

(iii) a generic Morse theorem, simultaneously for both Lloyd objectives and every N, when the density varies in the Banach manifold of strictly positive C^1 probability densities on a bounded convex domain.

Status relative to the open problem. The results below constitute *substantial partial progress*. They genuinely remove every definability assumption and apply to a residual set of non-globally-subanalytic densities. They do not prove convergence for every absolutely continuous target, for every positive C^1 density, for densities that vanish, or for arbitrary nonconvex supports. Accordingly, the unrestricted problem is not claimed to be solved.

2 Quantization objectives and Lloyd maps

Throughout the main theorem, $\Omega \subset \mathbb{R}^d$ is bounded, open, convex, and has Lipschitz boundary. We regard $C^1(\overline{\Omega})$ as the Banach space of C^1 functions on the closure, with its usual norm. Set

$$\mathcal{P}^1_+(\overline{\Omega}) = \left\{ f \in C^1(\overline{\Omega}) : \min_{\overline{\Omega}} f > 0, \ \int_\Omega f(x)\,\mathrm{d}x = 1 \right\}. \tag{1}$$

This is an open subset of the affine Banach hyperplane $\{f : \int_\Omega f = 1\}$ and hence is a Baire space. For $N \geq 1$, let

$$\mathcal{X}_N = \left\{ Y = (y_1, \dots, y_N) \in \Omega^N : y_i \neq y_j \text{ for } i \neq j \right\}. \tag{2}$$

2.1 Optimal quantization

For $f \in \mathcal{P}^1_+(\overline{\Omega})$ and $Y \in \mathcal{X}_N$, define

$$G_f(Y) = \frac{1}{2} \int_\Omega \min_{1 \leq i \leq N} |x - y_i|^2 f(x)\,\mathrm{d}x. \tag{3}$$

The open Voronoi cell of y_i relative to Ω is

$$V_i(Y) = \{x \in \Omega : |x - y_i| < |x - y_j| \text{ for all } j \neq i\}. \tag{4}$$

Since $y_i \in \Omega$ and the sites are distinct, $V_i(Y)$ has positive Lebesgue measure. Put

$$m_i(f, Y) = \int_{V_i(Y)} f(x)\,\mathrm{d}x, \qquad T_f(Y)_i = \frac{\int_{V_i(Y)} x f(x)\,\mathrm{d}x}{m_i(f, Y)}. \tag{5}$$

The optimal Lloyd iteration is $Y_{n+1} = T_f(Y_n)$.

2.2 Uniform quantization

Define

$$F_f(Y) = \frac{1}{2} W_2^2\left(f(x)\,\mathrm{d}x, \frac{1}{N} \sum_{i=1}^N \delta_{y_i} \right). \tag{6}$$

For $Y \in \mathcal{X}_N$ and $w \in \mathbb{R}^N$, let

$$L_i(Y, w) = \left\{x \in \Omega : |x - y_i|^2 - w_i \leq |x - y_j|^2 - w_j \text{ for all } j\right\}. \tag{7}$$

Introduce the concave dual functional

$$\Phi_f(Y, w) = \frac{1}{2} \int_\Omega \min_{1 \leq i \leq N} \big(|x - y_i|^2 - w_i\big) f(x)\, \mathrm{d}x + \frac{1}{2N} \sum_{i=1}^N w_i. \tag{8}$$

Semi-discrete Kantorovich duality [2, 5] gives $F_f(Y) = \max_{w \in \mathbb{R}^N} \Phi_f(Y, w)$ and a maximizing weight $w = w(f, Y)$, unique after imposing $w \cdot \mathbf{1} = 0$, such that

$$\int_{L_i(Y, w(f,Y))} f(x)\, \mathrm{d}x = \frac{1}{N} \qquad (i = 1, \ldots, N). \tag{9}$$

We abbreviate $L_i(f, Y) = L_i(Y, w(f, Y))$ and set

$$B_f(Y)_i = N \int_{L_i(f,Y)} x f(x)\, \mathrm{d}x. \tag{10}$$

The uniform Lloyd iteration is $Y_{n+1} = B_f(Y_n)$.

3 Main results

Theorem 3.1 (Generic Morse property and convergence)**.** *There exists a residual set $\mathcal{R} \subset \mathcal{P}^1_+(\overline{\Omega})$ such that the following holds. For every $f \in \mathcal{R}$ and every $N \geq 1$:*

(a) G_f and F_f are Morse functions on $\mathcal{X}_N$;

(b) every optimal Lloyd sequence $Y_{n+1} = T_f(Y_n)$ with $Y_0 \in \mathcal{X}_N$ converges to a nondegenerate critical point of G_f;

(c) every uniform Lloyd sequence $Y_{n+1} = B_f(Y_n)$ with $Y_0 \in \mathcal{X}_N$ converges to a nondegenerate critical point of F_f.

For either iteration, the trajectory has finite length and there are $C > 0$, $q \in (0, 1)$, and n_0 such that

$$\|Y_n - Y_\infty\| \leq C q^{n - n_0} \qquad (n \geq n_0).$$

The deterministic part of the argument is useful independently of genericity.

Theorem 3.2 (Critical-level criterion)**.** *Consider either Lloyd objective $E \in \{G_f, F_f\}$ and its corresponding Lloyd orbit (Y_n). Suppose the orbit is contained in a compact subset of the manifold on which E is C^1, and let*

$$E_\infty = \lim_{n \to \infty} E(Y_n).$$

If

$$\mathrm{Crit}(E) \cap \{E = E_\infty\} \tag{11}$$

is totally disconnected, then Y_n converges to a single critical point. It is enough that the critical points in the orbit closure be isolated.

If every point of the omega-limit set is a Morse–Bott critical point of E, then Y_n converges, the trajectory has finite length, and convergence is eventually geometric.

4 A general omega-limit principle

Lemma 4.1 (Connected omega-limit set). *Let (z_n) be a precompact sequence in a finite-dimensional normed space. If*

$$\|z_{n+1} - z_n\| \longrightarrow 0,$$

then its omega-limit set

$$\omega(z_0) = \bigcap_{m \geq 0} \overline{\{z_n : n \geq m\}}$$

is nonempty, compact, and connected.

Proof. Nonemptiness and compactness follow from precompactness. We first claim that

$$\operatorname{dist}(z_n, \omega(z_0)) \longrightarrow 0. \tag{12}$$

Otherwise, some subsequence would remain at distance at least $\varepsilon > 0$ from the omega-limit set. A convergent subsubsequence would have a limit in $\omega(z_0)$, contradicting that lower bound.

Suppose that $\omega(z_0)$ is disconnected. Since it is compact, it can be written as $A \cup B$, where A and B are nonempty disjoint compact sets. Put $\delta = \operatorname{dist}(A, B) > 0$ and choose $0 < \varepsilon < \delta/4$. The neighborhoods

$$U = \{z : \operatorname{dist}(z, A) < \varepsilon\}, \qquad V = \{z : \operatorname{dist}(z, B) < \varepsilon\}$$

are disjoint and satisfy $\operatorname{dist}(U, V) > \delta/2$. By (12), every sufficiently late iterate lies in $U \cup V$. Since the step lengths tend to zero, every sufficiently late step has length less than $\delta/2$. Thus, after some index, the sequence cannot pass from U to V or from V to U. One of A and B would then contain no accumulation point of the tail, a contradiction. Hence $\omega(z_0)$ is connected. □

Proposition 4.2 (Descent and relative error). *Let $U \subset \mathbb{R}^m$ be open, let $E \in C^1(U)$, and let $(z_n) \subset K \Subset U$, where K is compact. Suppose that there are constants $a, b > 0$ such that*

$$E(z_n) - E(z_{n+1}) \geq a\|z_{n+1} - z_n\|^2, \tag{13}$$

$$\|\nabla E(z_n)\| \leq b\|z_{n+1} - z_n\|. \tag{14}$$

Then $E(z_n)$ converges to a number E_∞ and

$$\omega(z_0) \subset \operatorname{Crit}(E) \cap \{E = E_\infty\}.$$

The omega-limit set is connected. Consequently, if the critical level set $\operatorname{Crit}(E) \cap \{E = E_\infty\}$ *is totally disconnected, then z_n converges.*

Proof. The energy is decreasing and bounded below on K, hence convergent. Summing (13) gives

$$\sum_{n=0}^{\infty} \|z_{n+1} - z_n\|^2 < \infty,$$

so the steps tend to zero. By (14), $\nabla E(z_n) \to 0$. Passing to a convergent subsequence and using continuity of E and ∇E proves the inclusion of the omega-limit set in the critical level. Connectedness follows from Lemma 4.1. A connected subset of a totally disconnected space is a singleton. A precompact sequence with a unique accumulation point converges to that point. □

Definition 4.3 (Morse–Bott point)**.** *A critical point $z_\star$ of a C^2 function E is Morse–Bott if, in a neighborhood of $z_\star$, the critical set is a C^2 submanifold M and*

$$\ker \nabla^2 E(z_\star) = T_{z_\star} M.$$

A Morse critical point is the special case in which M is zero-dimensional.

Lemma 4.4 (Morse–Bott gradient inequality)**.** *If $z_\star$ is a Morse–Bott critical point of $E \in C^2(U)$, then there are a neighborhood Q of $z_\star$ and $C > 0$ such that*

$$|E(z) - E(z_\star)|^{1/2} \le C\|\nabla E(z)\| \qquad (z \in Q). \tag{15}$$

Proof. The Morse–Bott lemma gives local coordinates (u, v) in which the critical manifold is $\{v = 0\}$ and

$$E(u, v) = E(z_\star) + \frac{1}{2}\|v_+\|^2 - \frac{1}{2}\|v_-\|^2.$$

In these coordinates, $|E - E(z_\star)|^{1/2} \le C_1\|v\|$, while the normal component of the gradient is bounded below by $C_2\|v\|$. The coordinate map and its inverse have bounded derivatives after shrinking the neighborhood, which yields (15) in the original coordinates. □

Proposition 4.5 (Morse–Bott convergence and rate)**.** *Under the hypotheses of Proposition 4.2, suppose that every point of $\omega(z_0)$ is Morse–Bott. Then z_n converges to a point z_∞, the trajectory has finite length, and convergence is eventually geometric.*

Proof. Every point of the compact set $\omega(z_0)$ has energy E_∞. A finite subcover of the neighborhoods in Lemma 4.4 gives a neighborhood Q of $\omega(z_0)$ and a constant $C > 0$ such that

$$(E(z) - E_\infty)^{1/2} \le C\|\nabla E(z)\| \tag{16}$$

whenever $z \in Q$ and $E(z) \ge E_\infty$. By (12), the iterates eventually lie in Q.

Set $e_n = E(z_n) - E_\infty$. For all sufficiently large n, (14) and (16) imply

$$e_n^{1/2} \le Cb\|z_{n+1} - z_n\|.$$

Together with (13), this gives

$$e_n - e_{n+1} \ge \frac{a}{C^2 b^2} e_n. \tag{17}$$

If the coefficient on the right is at least one, (17) forces the sequence to reach energy E_∞ and then become stationary. Otherwise, $e_{n+1} \le \rho e_n$ for some $\rho \in (0, 1)$. Moreover, (13) gives

$$\|z_{n+1} - z_n\| \le a^{-1/2} e_n^{1/2}.$$

Hence the step lengths are summable, and the tail beginning at n is $O(\rho^{n/2})$. Thus (z_n) is Cauchy, converges, has finite length, and converges geometrically. □

5 Descent and compactness for Lloyd iterations

5.1 A quantitative barycenter lemma

The following statement includes explicitly the probability normalization used in its proof.

Lemma 5.1 (Barycenters stay uniformly inside)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and convex, let $D = \operatorname{diam}(\Omega)$, and let f satisfy*

$$0 < m \le f(x) \le M < \infty \quad (x \in \Omega), \qquad \int_\Omega f(x)\,\mathrm{d}x = 1.$$

For every $a > 0$ there is $\delta = \delta(a, m, M, D, d) > 0$ such that the following holds. If $C \subset \Omega$ is convex and $\int_C f \ge a$, then its f-barycenter

$$b_C = \frac{\int_C x f(x)\,\mathrm{d}x}{\int_C f(x)\,\mathrm{d}x}$$

satisfies $\operatorname{dist}(b_C, \partial C) \ge \delta$*. One possible choice is*

$$\delta = \frac{ma^2}{4M^2 A_d}, \qquad A_d = \omega_{d-1} D^{d-1}, \tag{18}$$

with $A_1 = 1$.

Proof. Replacing C by its closure does not change its mass or barycenter. Since C has positive mass and $f \le M$, it has positive Lebesgue measure and hence nonempty interior. Strict positivity of f then places b_C in $\operatorname{int} C$. Let $h_C(u) = \sup_{x \in C} u \cdot x$ be its support function. For $b_C \in \operatorname{int} C$,

$$r := \operatorname{dist}(b_C, \partial C) = \min_{|u|=1} \big(h_C(u) - u \cdot b_C\big).$$

Choose a minimizing unit vector u and put $t(x) = h_C(u) - u \cdot x$. Then $0 \le t \le D$ on C and $t(b_C) = r$.

Let $v = |C|$. Since $f \le M$ and $\int_C f \ge a$, we have $v \ge a/M$. Every section of C perpendicular to u has $(d-1)$-dimensional measure at most A_d. Therefore the layer $\{x \in C : 0 \le t(x) \le s\}$ has volume at most $A_d s$. With $s = v/(2A_d)$, at least half of the volume of C lies in $\{t \ge s\}$. Consequently,

$$\int_C t(x) f(x)\,\mathrm{d}x \ge m\, s\, \frac{v}{2} = \frac{mv^2}{4A_d} \ge \frac{ma^2}{4M^2 A_d}.$$

Because t is affine,

$$r = t(b_C) = \frac{\int_C t(x) f(x)\,\mathrm{d}x}{\int_C f(x)\,\mathrm{d}x}.$$

The denominator is at most 1, so (18) follows. □

5.2 Optimal quantization

For $Y \in \mathcal{X}_N$, the standard first-variation identity is

$$\nabla_{y_i} G_f(Y) = \int_{V_i(Y)} (y_i - x) f(x)\,\mathrm{d}x = m_i(f, Y)\big(y_i - T_f(Y)_i\big). \tag{19}$$

The centroid identity and the use of the old Voronoi partition as a competitor for the new sites give

$$G_f(Y) - G_f(T_f(Y)) \ge \frac{1}{2} \sum_{i=1}^N m_i(f, Y) \|T_f(Y)_i - y_i\|^2. \tag{20}$$

We use the classical nondegeneracy theorem for the optimal Lloyd orbit. Under the present hypotheses, Corollary 3.7 of Emelianenko, Ju, and Rand gives an orbit-dependent constant $\ell > 0$ such that

$$\inf_{n\geq 0} \min_{1\leq i\leq N} m_i(f, Y_n) \geq \ell; \tag{21}$$

see also Proposition 3.4 of [5]. Thus

$$G_f(Y_n) - G_f(Y_{n+1}) \geq \frac{\ell}{2}\|Y_{n+1} - Y_n\|^2. \tag{22}$$

Since every cell mass is at most one, (19) also gives

$$\|\nabla G_f(Y_n)\| \leq \|Y_{n+1} - Y_n\|. \tag{23}$$

Let $m = \min_{\overline{\Omega}} f$ and $M = \max_{\overline{\Omega}} f$. Applying Lemma 5.1 to each Voronoi cell, with $a = \ell$, shows that every Y_{n+1} is uniformly separated from $\partial\Omega$ and from every boundary of its own Voronoi cell. If two updated sites were less than twice that separation apart, the corresponding interior balls would overlap, contradicting disjointness of the Voronoi interiors. Hence, after one step, the orbit lies in a compact subset of $\mathcal{X}_N$.

5.3 Uniform quantization

The envelope theorem for the semi-discrete dual yields [5]

$$\nabla_{y_i} F_f(Y) = \frac{1}{N}\big(y_i - B_f(Y)_i\big) = \int_{L_i(f,Y)} (y_i - x) f(x)\,\mathrm{d}x. \tag{24}$$

Using the old Laguerre cells as a transport competitor after moving each site to its cell barycenter gives [5]

$$F_f(Y) - F_f(B_f(Y)) \geq \frac{1}{2N}\|B_f(Y) - Y\|^2. \tag{25}$$

Thus (13)–(14) hold with

$$a = \frac{1}{2N}, \qquad b = \frac{1}{N}. \tag{26}$$

Every balanced Laguerre cell has mass $1/N$ and is convex. Applying Lemma 5.1 with $a = 1/N$ shows that every point of $B_f(Y)$ is uniformly separated from $\partial\Omega$ and from the boundary of its own Laguerre cell. The same disjoint-interior argument gives a uniform separation between distinct barycenters. Consequently, after one step, every uniform Lloyd orbit lies in a fixed compact subset of $\mathcal{X}_N$.

Proof of Theorem 3.2. For optimal quantization, (22)–(23) and the preceding compactness argument verify the hypotheses of Proposition 4.2. For uniform quantization, the same follows from (24)–(26). The totally disconnected conclusion follows from Proposition 4.2, and the Morse–Bott conclusion follows from Proposition 4.5. □

6 Joint second-order regularity

The genericity argument requires C^1 dependence of the gradients on both the density and the sites. We give the details needed to pass from the standard fixed-density formulas to joint regularity.

For distinct sites Y and weights w, define

$$M_i(f, Y, w) = \int_{L_i(Y,w)} f(x)\,\mathrm{d}x, \qquad P_i(f, Y, w) = \int_{L_i(Y,w)} x f(x)\,\mathrm{d}x.$$

Let $\mathcal{U}_N$ be the open set of pairs (Y, w) for which Y has distinct components and every Laguerre cell has positive Lebesgue measure.

Lemma 6.1 (Joint regularity of cell masses and moments). *The maps*

$$(f, Y, w) \longmapsto M_i(f, Y, w), \qquad (f, Y, w) \longmapsto P_i(f, Y, w)$$

are C^1 from $C^1(\overline{\Omega}) \times \mathcal{U}_N$ to $\mathbb{R}$ and $\mathbb{R}^d$, respectively.

Proof. For a fixed $f \in C^1(\overline{\Omega})$, the Euclidean second-order regularity theorem of de Gournay, Kahn, and Lebrat applies on a bounded convex Lipschitz domain whenever the sites are distinct and all cells have positive volume [2]. The dual cell integral is C^2 in (Y, w); its weight derivatives are the cell masses, while its site derivatives are $y_i M_i - P_i$. Hence both M_i and P_i are C^1 in (Y, w). Their first derivatives are finite sums of volume terms and facet integrals. For example, the derivative of a cell mass in a weight direction is a weighted graph-Laplacian expression with coefficients

$$a_{ij}(f, Y, w) = \int_{\partial L_i \cap \partial L_j} \frac{f(x)}{2|y_i - y_j|}\,\mathrm{d}\mathcal{H}^{d-1}(x) \qquad (i \neq j), \tag{27}$$

with the convention that the integral is zero when the common facet has zero $(d-1)$-measure. The corresponding site derivatives have the same form with affine factors in x.

Dependence on f is linear at fixed cells:

$$D_f M_i(f, Y, w)[h] = \int_{L_i(Y,w)} h(x)\,\mathrm{d}x, \qquad D_f P_i(f, Y, w)[h] = \int_{L_i(Y,w)} x h(x)\,\mathrm{d}x.$$

As (Y, w) varies in $\mathcal{U}_N$, the defining affine hyperplanes vary continuously and the cell boundaries have Lebesgue measure zero. Hence the cells converge in measure locally, and the two displayed operators vary continuously in operator norm on $C^1(\overline{\Omega})$, because $\|h\|_\infty \leq \|h\|_{C^1}$. Finally, the volume and facet formulas for derivatives in (Y, w) are linear in f and depend continuously on f in the uniform norm. The continuity statement in the cited second-order theorem handles changes of cell combinatorics. These facts show that the full Fréchet derivative with respect to (f, Y, w) exists and is continuous. $\square$

Proposition 6.2 (Regularity of the universal gradients). *For every N, the maps*

$$\begin{aligned} \Gamma_N^{\mathrm{opt}} &: \mathcal{P}_+^1(\overline{\Omega}) \times \mathcal{X}_N \longrightarrow (\mathbb{R}^d)^N, & \Gamma_N^{\mathrm{opt}}(f, Y) &= \nabla G_f(Y), \\ \Gamma_N^{\mathrm{unif}} &: \mathcal{P}_+^1(\overline{\Omega}) \times \mathcal{X}_N \longrightarrow (\mathbb{R}^d)^N, & \Gamma_N^{\mathrm{unif}}(f, Y) &= \nabla F_f(Y) \end{aligned}$$

are C^1. In particular, G_f and F_f are C^2 on $\mathcal{X}_N$ for each fixed f.

Proof. For optimal quantization, the weights are fixed at zero and every Voronoi cell has positive volume. By Lemma 6.1,

$$\Gamma_{N,i}^{\mathrm{opt}}(f, Y) = y_i M_i(f, Y, 0) - P_i(f, Y, 0)$$

is jointly C^1.

For uniform quantization, consider the mass-balance map on the gauge space $\mathbf{1}^\perp$,

$$\mathcal{M}(f,Y,w) = \left(M_i(f,Y,w) - \frac{1}{N}\right)_{i=1}^{N} \in \mathbf{1}^\perp.$$

It is C^1 by Lemma 6.1. At a balanced weight, all cells have positive volume. The derivative $D_w\mathcal{M}$ is the weighted graph Laplacian determined by (27); more precisely,

$$v^\top D_w\mathcal{M}(f,Y,w)v = \sum_{1\le i<j\le N} a_{ij}(f,Y,w)(v_i - v_j)^2.$$

The graph whose edges correspond to positive-area common facets is connected. Indeed, choose interior points in any two cells and perturb the line segment joining them so that it avoids the codimension-two skeleton of the finite polyhedral partition; the segment then crosses a chain of codimension-one interfaces. Since $f > 0$, every coefficient on such an interface is positive. Therefore the kernel of $D_w\mathcal{M}$ is exactly $\operatorname{span}\{\mathbf{1}\}$, and its restriction to $\mathbf{1}^\perp$ is an isomorphism.

The Banach-space implicit-function theorem now gives a local C^1 map $w = w(f,Y) \in \mathbf{1}^\perp$ solving (9). The normalized solution is unique, so the local maps agree on overlaps and define a global C^1 map. By Lemma 6.1,

$$\Gamma^{\mathrm{unif}}_{N,i}(f,Y) = \frac{1}{N}y_i - P_i(f,Y,w(f,Y))$$

is jointly C^1. □

7 Cellwise density perturbations and submersivity

The tangent space of $\mathcal{P}^1_+(\overline{\Omega})$ is

$$\mathcal{H} = \left\{h \in C^1(\overline{\Omega}) : \int_\Omega h(x)\,\mathrm{d}x = 0\right\}. \tag{28}$$

Proposition 7.1 (Cellwise controllability). *For every N, every $(f,Y) \in \mathcal{P}^1_+(\overline{\Omega}) \times \mathcal{X}_N$, and either universal gradient map*

$$\Gamma_N \in \{\Gamma^{\mathrm{opt}}_N, \Gamma^{\mathrm{unif}}_N\},$$

the partial derivative

$$D_f\Gamma_N(f,Y) : \mathcal{H} \longrightarrow (\mathbb{R}^d)^N$$

is surjective. Consequently, both universal gradient maps are C^1 submersions.

Proof. Fix $q = (q_1,\ldots,q_N) \in (\mathbb{R}^d)^N$.

Optimal quantization. Choose pairwise disjoint open balls $U_i \Subset V_i(Y)$. Let $\psi_i \in C^\infty_c(U_i)$ satisfy $\int \psi_i = 1$, and define

$$h_i(x) = q_i\cdot\nabla\psi_i(x), \qquad h = \sum_{i=1}^N h_i. \tag{29}$$

Each h_i has integral zero, so $h \in \mathcal{H}$. The Voronoi cells do not depend on f, and therefore

$$\begin{aligned} D_f\Gamma^{\mathrm{opt}}_{N,i}(f,Y)[h] &= \int_{V_i(Y)} (y_i - x)h(x)\,\mathrm{d}x \\ &= \int_{U_i} (y_i - x)(q_i\cdot\nabla\psi_i(x))\,\mathrm{d}x = q_i. \end{aligned}$$

The final equality follows componentwise by integration by parts; no boundary term occurs because ψ_i is compactly supported in U_i.

Uniform quantization. Let $w = w(f, Y)$ and $L_i = L_i(f, Y)$. Each L_i has positive mass and $f > 0$, so it has nonempty interior. Choose pairwise disjoint balls $U_i \Subset \operatorname{int} L_i$ and define h by (29). For all sufficiently small t, the density $f_t = f + th$ remains positive and normalized. Moreover,

$$\int_{L_i} f_t = \frac{1}{N} + t \int_{L_i} h = \frac{1}{N},$$

because only h_i is supported in L_i and $\int h_i = 0$.

For fixed Y, the semi-discrete dual objective is concave in w, and its first-order condition is exactly the mass-balance condition. Thus the same weight w is an optimal dual weight for f_t. Since the cells are determined only by (Y, w), they remain equal to L_i along this perturbation. Hence

$$D_f \Gamma^{\mathrm{unif}}_{N,i}(f, Y)[h] = \int_{L_i} (y_i - x) h(x) \, \mathrm{d}x = q_i$$

by the same integration-by-parts computation. Since q was arbitrary, both partial derivatives are surjective. $\square$

Remark 7.2. *The zero-mass derivative $q_i \cdot \nabla \psi_i$ has exactly the two required features. It preserves total mass—and, in the uniform problem, each individual balanced cell mass—while its first moment can be prescribed arbitrarily. This local controllability is the essential input to transversality.*

8 Parametric transversality and generic Morse functions

We record a version of parametric transversality whose proof uses only finite-dimensional Sard theory; compare [1].

Theorem 8.1 (Parametric transversality)**.** *Let P be a Baire C^1 Banach manifold, let X be a second-countable m-dimensional C^1 manifold, and let*

$$\Phi : P \times X \longrightarrow \mathbb{R}^m$$

be C^1. Suppose that $D_p\Phi(p, x) : T_pP \to \mathbb{R}^m$ is surjective for every (p, x). Then

$$\{p \in P : 0 \textit{ is a regular value of } \Phi(p, \cdot)\}$$

is residual in P.

Proof. Let $K_1 \subset K_2 \subset \cdots$ be a compact exhaustion of X. For a compact $K \subset X$, let $\mathcal{A}_K$ be the set of parameters p for which $\Phi(p, \cdot)$ is transverse to zero at every zero lying in K.

First, $\mathcal{A}_K$ is open. Otherwise there would be $p_j \to p \in \mathcal{A}_K$ and $x_j \in K$ such that $\Phi(p_j, x_j) = 0$ and $D_x\Phi(p_j, x_j)$ is not surjective. Passing to a subsequence, $x_j \to x \in K$. Continuity gives $\Phi(p, x) = 0$, and the set of nonsurjective linear maps is closed, so $D_x\Phi(p, x)$ is not surjective, a contradiction.

To prove density, fix $p_0 \in P$. For each $x \in K$, surjectivity of $D_p\Phi(p_0, x)$ provides finitely many tangent directions whose images span $\mathbb{R}^m$. By continuity, the same directions work in a neighborhood of x. A finite subcover of K yields a finite-dimensional subspace $H \subset T_{p_0}P$ such that the restriction of the parameter derivative to H is surjective near $\{p_0\} \times K$. In a local chart of P, consider

$$\Psi : H \times O \longrightarrow \mathbb{R}^m, \qquad \Psi(a, x) = \Phi(p_0 + a, x),$$

where O is an open neighborhood of K and a is sufficiently small. The map Ψ is a submersion.

Consequently, $S = \Psi^{-1}(0)$ is a C^1 manifold of dimension $\dim H$. Let $\pi : S \to H$ be the projection. A parameter a is a regular value of π if and only if the slice $x \mapsto \Psi(a, x)$ is transverse to zero: this follows directly from

$$T_{(a,x)}S = \{(u, v) : D_a\Psi(a, x)u + D_x\Psi(a, x)v = 0\}$$

and the surjectivity of $D_a\Psi$. Since the domain and target of π have the same dimension, the C^1 Sard theorem applies and regular values are dense. Thus arbitrarily small perturbations of p_0 belong to $\mathcal{A}_K$. Hence each $\mathcal{A}_{K_r}$ is open and dense, and $\bigcap_r \mathcal{A}_{K_r}$ is the desired residual set. □

Proposition 8.2 (Generic Morse objectives for fixed N). *For every fixed N, the sets*

$$\mathcal{R}_N^{\mathrm{opt}} = \{f \in \mathcal{P}_+^1(\overline{\Omega}) : G_f \text{ is Morse on } \mathcal{X}_N\},$$
$$\mathcal{R}_N^{\mathrm{unif}} = \{f \in \mathcal{P}_+^1(\overline{\Omega}) : F_f \text{ is Morse on } \mathcal{X}_N\}$$

are residual.

Proof. Apply Theorem 8.1 to the universal gradient maps in Proposition 6.2. The parameter derivative is surjective by Proposition 7.1. Since $\dim \mathcal{X}_N = Nd = \dim(\mathbb{R}^d)^N$, transversality of the slice $Y \mapsto \Gamma_N(f, Y)$ to zero says that, at every zero,

$$D_Y\Gamma_N(f, Y) : (\mathbb{R}^d)^N \longrightarrow (\mathbb{R}^d)^N$$

is an isomorphism. This derivative is the Hessian of the corresponding objective. Thus every critical point is nondegenerate. □

Proof of Theorem 3.1. Set

$$\mathcal{R} = \bigcap_{N=1}^{\infty} \left(\mathcal{R}_N^{\mathrm{opt}} \cap \mathcal{R}_N^{\mathrm{unif}}\right). \tag{30}$$

This is residual in the Baire space $\mathcal{P}_+^1(\overline{\Omega})$. Fix $f \in \mathcal{R}$ and N. Both objectives are Morse, hence their critical sets are discrete and therefore totally disconnected. The compactness and descent estimates established above verify the hypotheses of Proposition 4.2; that proposition gives convergence of every orbit. Since the limiting critical point is Morse, Proposition 4.5 gives finite length and eventual geometric convergence. □

9 The result is genuinely beyond global subanalyticity

For completeness, we show that the globally subanalytic densities form a small subset of the ambient C^1 space. This also prevents the generic theorem from being merely a reformulation of the definable case.

Let $\mathcal{S} \subset \mathcal{P}_+^1(\overline{\Omega})$ be the set of densities admitting a globally subanalytic representative on $\mathbb{R}^d$. Analytic stratification of globally subanalytic graphs implies that every $f \in \mathcal{S}$ is real analytic on some nonempty open ball contained in Ω [6].

Proposition 9.1 (Definable densities are meagre). *The set $\mathcal{S}$ is meagre in $\mathcal{P}_+^1(\overline{\Omega})$. Consequently, $\mathcal{R} \setminus \mathcal{S}$ is residual, where $\mathcal{R}$ is the set in* (30).

Proof. Let K range over closed balls with rational center and radius such that $K \Subset \Omega$, and let $k \in \mathbb{N}$. Define

$$\mathcal{L}_{K,k} = \left\{ f \in \mathcal{P}^1_+(\overline{\Omega}) : |Df(x) - Df(y)| \le k|x-y| \text{ for all } x, y \in K \right\}.$$

Each $\mathcal{L}_{K,k}$ is closed in the C^1 topology. It has empty interior. Indeed, inside K choose a small ball and add an arbitrarily small compactly supported C^1 cusp of the form $|x_1|^{3/2}$ times a smooth cutoff. Subtract a smooth correction supported in a disjoint ball so that the perturbation has integral zero. After scaling, positivity and normalization are preserved, but the derivative is not Lipschitz on K. Thus no C^1 neighborhood is contained in $\mathcal{L}_{K,k}$.

If $f \in \mathcal{S}$, then f is analytic on some open ball. On a smaller rational closed ball K, the Hessian is bounded, so Df is k-Lipschitz for some integer k. Therefore

$$\mathcal{S} \subset \bigcup_{K,k} \mathcal{L}_{K,k},$$

a countable union of nowhere dense sets. Hence $\mathcal{S}$ is meagre. The complement of $\mathcal{S}$ is residual, so its intersection with $\mathcal{R}$ is residual. $\square$

Remark 9.2. *The preceding proposition is a category statement, not an explicit membership test for $\mathcal{R}$. For example, smooth positive densities containing a localized oscillatory term $e^{-1/t^2}\sin(1/t)$ are not globally subanalytic, but the theorem does not assert that every such explicitly written density is Morse.*

10 Scope and conclusion

The argument identifies the precise obstruction left after the standard descent analysis: a Lloyd orbit can fail to have a unique limit only if its connected omega-limit set lies in a connected continuum of critical points at one energy level. Total disconnectedness rules this out. The cellwise perturbation lemma then shows that critical continua are nongeneric with respect to the density, because the universal gradient maps are submersions.

The resulting theorem is strong within its regime: one residual set of densities works simultaneously for every N, for both optimal and uniform quantization, and for every distinct initialization. It also yields a local geometric rate. Nevertheless, it remains a partial resolution. Extending the conclusion from residual densities to all positive C^1 densities, allowing vanishing or rough densities, and treating general nonconvex supports are separate questions not settled here.

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